

SkyMetric

India Airfare Price Index System

Problem Statement & Proposed Solution

Real-Time DGCA Passenger-Weighted Airfare Index

Tech Stack: Python 3.14 | FastAPI | Playwright | Next.js 16 | Shadcn UI | Recharts

Database: SQLite + SQLAlchemy ORM

Index Formulas: Jevons | Laspeyres | Chain-Laspeyres

1. Problem Statement

Background

India is the third-largest domestic aviation market in the world, with over 150 million passengers flying annually (DGCA 2023-24). Despite this scale, there is no publicly available, standardized, real-time airfare price index for the Indian market.

Core Problems Identified

1. **No Centralized Fare Index:** The Directorate General of Civil Aviation (DGCA) publishes passenger volume statistics and airline performance data, but does not compute or publish a fare price index. Passengers and analysts lack a benchmark to understand whether airfares are rising, falling, or stable.
2. **Opaque Pricing by OTAs:** Online Travel Agencies (MakeMyTrip, Yatra, EaseMyTrip) and airline websites (IndiGo, Air India) display point-in-time fares with no historical context. A passenger booking a DEL-BOM ticket for Rs.5,200 has no way to know if this is above or below the market average for that corridor.
3. **No Advance-Purchase Analysis:** Airline pricing is heavily influenced by booking lead time (how far in advance the ticket is purchased). There is no public tool that quantifies the price elasticity across booking windows (T+1 vs T+7 vs T+30 vs T+45).
4. **Missing Sector-Level Granularity:** National-level inflation indices (CPI, WPI) do not capture corridor-specific fare dynamics. The DEL-IXS (Silchar) route behaves very differently from DEL-BOM (Mumbai), but both are treated identically in aggregate data.
5. **No Carrier Comparison Framework:** Passengers cannot easily compare fare levels across carriers on the same route. Market share data exists (IndiGo at 63.6%) but fare competitiveness data is not publicly available.
6. **No Structured Data for Policymakers:** Organizations like MoSPI, RBI, and NSO need structured airfare data for inflation modeling and monetary policy decisions. Currently, no API or dataset provides this in a machine-readable format.

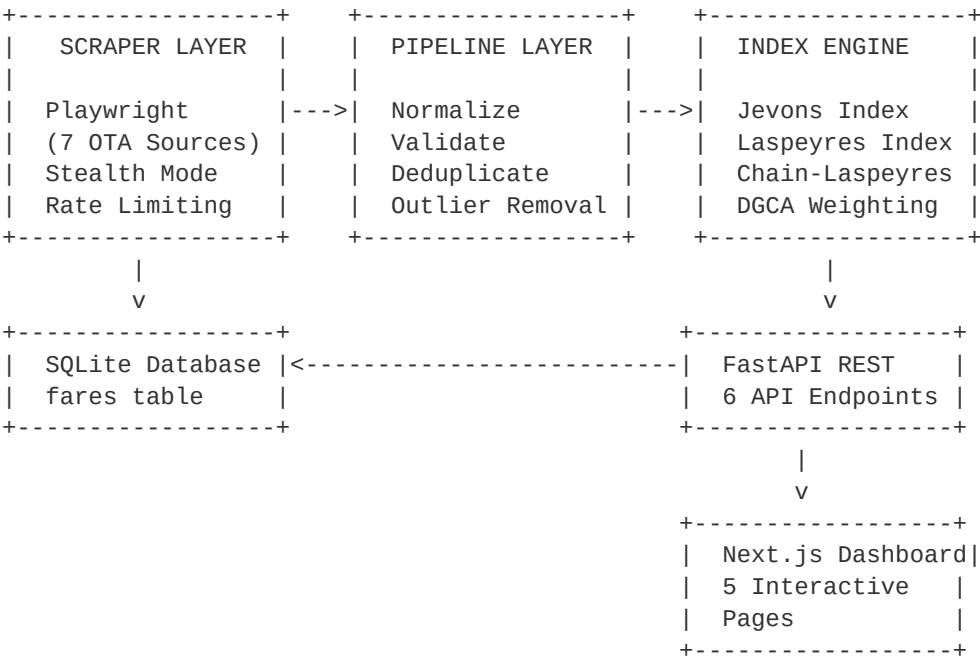
Impact

The absence of a fare index leads to: (a) passengers overpaying due to lack of price discovery, (b) airlines unable to benchmark against competitors, (c) policymakers making inflation estimates without real airfare data, and (d) researchers lacking a reliable dataset for aviation economics studies.

2. Proposed Solution: SkyMetric

SkyMetric (SkyMetric Airfare Index for India) is an end-to-end system that scrapes real-time fare data from multiple airline and OTA sources, computes a DGCA Passenger-Weighted Basket Index, and presents it through a modern web dashboard with interactive visualizations.

System Architecture



Key Design Decisions

- DGCA Passenger-Weighted Basket: Corridor weights are derived from DGCA city-pair passenger traffic statistics (2023-24), ensuring the index reflects actual passenger distribution.
- 10 Monitored Corridors: 8 Metro Trunk routes (DEL-BOM, DEL-BLR, BOM-BLR, etc.) + 2 Regional Thin routes (DEL-IXS, DEL-DHM) covering ~90% of domestic passenger traffic.
- 6 Carriers Tracked: IndiGo (63.6%), Air India (15.2%), Vistara (9.1%), SpiceJet (5.5%), Akasa Air (4.1%), Air India Express (2.5%).
- 5 Advance-Purchase Windows: T+1 (Emergency), T+7 (Short-Horizon), T+15 (Anchor), T+30 (Baseline), T+45 (Early Bird) to capture booking lead-time elasticity.
- Real-Time Scraping: Playwright headless browser with stealth mode, random user agents, rate limiting (2-3s delays), and exponential backoff for reliability.

3. Index Methodology

Jevons Index (Elementary)

The Jevons index is the geometric mean of price relatives. It is used as the elementary index when no explicit quantity weights are available.

$$I = (\text{product}(p_t / p_{t0}))^{(1/n)} * 100$$

Laspeyres Index (Weighted)

The Laspeyres index uses DGCA passenger weights to compute a weighted arithmetic mean of price relatives across corridors.

$$I = (\text{sum}(w_i * p_{ti} / p_{t0i}) / \text{sum}(w_i)) * 100$$

Chain-Laspeyres (Rolling)

The Chain-Laspeyres index links period-to-period indices multiplicatively, allowing the basket composition to change over time.

$$I_{\text{chain}} = 100 * \text{product}(\text{link}_i / 100)$$

National Index Computation

The national SkyMetric headline index is computed at the T+15 advance window (the anchor window). Sector indices for each corridor are computed using the Jevons formula, then aggregated using DGCA passenger-weighted averages to produce the national index. Base value is set to 100.00.

The system also computes sub-indices for all 5 advance windows (T+1, T+7, T+15, T+30, T+45), allowing users to see how the index varies by booking lead time.

Data Cleaning Pipeline

1. Validation: Require origin, destination, carrier, and total_fare > 0.
2. Normalization: Standardize IATA codes, recompute fares, fill defaults.
3. Deduplication: Remove cross-platform duplicates, keeping latest timestamp.
4. Outlier Removal: Z-score threshold of 3.0 to filter anomalous fares.

4. Dashboard Features

Page 1: National Airfare Price Index (Home)

- KPI Cards: Headline index, 24h change (%), monitored corridors count, pipeline status.
- 30-Day Line Chart: National price index trend with Base=100 reference line.
- Advance Booking Sub-Indices: T+1 to T+45 cards showing per-window index values.
- Corridor Index Breakdown: Table of 10 corridors with index, change %, classification.
- Fare Breakdown Donut: Average fare composition (base fare, taxes, convenience, UDF).
- Day-of-Week Heatmap: Which days are cheapest vs most expensive per corridor.
- Scraper Status Dashboard: Run scraper, view quotes collected, sources checked.

Page 2: Route Heatmap

- Sector x Date fare matrix for 10 corridors over 30 days.
- Origin/Destination city filter dropdowns.
- Color-coded cells: Emerald (Low) -> Yellow (Medium) -> Orange (High) -> Red (Peak).
- Row averages and summary stats: Lowest, Highest, Average Fare, Fare Spread (%).
- Interactive tooltips with exact fare values on hover.

Page 3: Advance Purchase Elasticity

- Multi-line chart showing lead-time price curves per corridor.
- X-axis: T+1 to T+45 advance windows. Y-axis: Average fare in INR.
- 5 summary cards: Emergency (T+1), Short-Horizon (T+7), Anchor (T+15), Baseline (T+30), Early Bird (T+45).

Page 4: Carrier Analysis

- Market Share Distribution: Horizontal bar chart for 6 carriers.
- Average Base Fare: Vertical bar chart comparing carrier pricing.
- Carrier Detail Table: Market share, avg fare, taxes (18%), total estimated fare.

Page 5: API Explorer

- Embedded FastAPI Swagger UI for interactive API testing.
- Reference grid of 6 GET endpoints for programmatic access.

5. Technical Stack

Backend

Component	Technology	Purpose
Language	Python 3.14	Core application logic
Web Framework	FastAPI	REST API with async support
Database	SQLite + SQLAlchemy	Fare storage and querying
Scraper	Playwright	Headless browser automation
Data Validation	Pydantic Settings	Configuration management
Testing	pytest (45/45 passing)	Unit and integration tests

Frontend

Component	Technology	Purpose
Framework	Next.js 16.3.4	React SSR/SSG with Turbopack
Language	TypeScript	Type-safe development
UI Library	Shadcn UI + Tailwind CSS	Modern component system
Charts	Recharts	Interactive data visualizations
State	TanStack Query	Server state caching/polling
Icons	Lucide React	Consistent icon system

6. Future Scope

- Real-Time Streaming: Replace periodic scraping with WebSocket-based live fare feeds.
- Airline API Integration: Direct API access to IndiGo, Air India for structured data.
- ML Price Forecasting: LSTM/Transformer models for fare prediction at T-7, T-15, T-30.
- Consumer Alerts: Push notifications when fares drop below user-defined thresholds.
- International Expansion: Extend to SAARC and ASEAN corridors.
- MoSPI/RBI Integration: Official API for inflation modeling and monetary policy.
- Mobile App: React Native companion for on-the-go fare tracking.
- Historical Database: Build a multi-year fare archive for trend analysis.